

MAT2001

Statistics for Engineers

Module 3

Correlation and Regression

Syllabus

Correlation and regression:

Correlation and Regression – Partial and Multiple correlation- Multiple regression.

Covariance

$$\text{Var}(X) = E[(X - E(X)) \cdot (X - E(X))]$$

$$\text{Covar}(X, Y) = E[(X - E(X)) \cdot (Y - E(Y))]$$

Correlation

CORRELATION COEFFICIENT

As the variance $E\{X - E(X)\}^2$ measures the variations of the R.V. X from its mean value $E(X)$, the quantity $E\{[X - E(X)][Y - E(Y)]\}$ measures the simultaneous variation of two R.V.'s X and Y from their respective means and hence it is called the *covariance* of X , Y and denoted as $\text{Cov}(X, Y)$.

$\text{Cov}(X, Y) = E\{[X - E(X)][Y - E(Y)]\}$ is also called the *product moment* of X and Y and is also denoted as $p(X, Y)$.

$\frac{p(x, y)}{\sigma_x \sigma_y}$ is a measure of intensity of linear relationship between X and Y and is called *Karl-Pearson's Product Moment Correlation Coefficient* or simply *correlation coefficient* between X and Y . It is denoted by $r(X, Y)$ or r_{XY} or simply r .

Thus

$$r_{XY} = \frac{E\{[X - E(X)][Y - E(Y)]\}}{\sqrt{E\{X - E(X)\}^2 E\{Y - E(Y)\}^2}} \quad (1)$$

since σ_x , the standard deviation of X is the positive square root of the variance of X .

$$r_{XY} = \frac{E\{[X - E(X)][Y - E(Y)]\}}{\sqrt{E\{X - E(X)\}^2 E\{Y - E(Y)\}^2}}$$

$$r_{XY} = \frac{E(XY) - E(X) \cdot E(Y)}{\sqrt{\{E(X^2) - E^2(X)\} \{E(Y^2) - E^2(Y)\}}}$$

$$r_{XY} = \frac{\frac{1}{n} \sum x_i y_i - \frac{1}{n} \sum x_i \cdot \frac{1}{n} \sum y_i}{\sqrt{\left\{ \frac{1}{n} \sum x_i^2 - \left(\frac{1}{n} \sum x_i \right)^2 \right\} \left\{ \frac{1}{n} \sum y_i^2 - \left(\frac{1}{n} \sum y_i \right)^2 \right\}}}$$

$$r_{XY} = \frac{n \sum xy - \sum x \cdot \sum y}{\sqrt{\{n \sum x^2 - (\sum x)^2\} \{n \sum y^2 - (\sum y)^2\}}}$$

Properties of Correlation Coefficient

1. $-1 \leq r_{XY} \leq 1$ or $|\text{Cov}(X, Y)| \leq \sigma_X \cdot \sigma_Y$.

Note: When $0 < r_{XY} \leq 1$, the correlation between X and Y is said to be *positive* or *direct*.

When $-1 \leq r_{XY} \leq 0$, the correlation is said to be *negative* or *inverse*.

When $-1 \leq r_{XY} \leq -0.5$ or $0.5 \leq r_{XY} \leq 1$, the correlation is assumed to be high, otherwise the correlation is assumed to be poor.

2. Correlation coefficient is independent of change of origin and scale.

Example:

Compute the coefficients of correlation between X and Y using the following data:

X :	65	67	66	71	67	70	68	69
Y :	67	68	68	70	64	67	72	70

Comment about the nature of correlation.

Solution:

We effect change of origin in respect of both X and Y . The new origins are chosen at or near the average of extreme values. Thus we take $\frac{65+71}{2} = 68$ as the new origin for X and $\frac{64+72}{2} = 68$ as the new origin for Y . viz., we put $u_i = (x_i - 68)$ and $v_i = y_i - 68$ and find r_{UV} .

$X = x_i$	$Y = y_i$	$u_i = x_i - 68$	$v_i = y_i - 68$	u_i^2	v_i^2	$u_i v_i$
65	67	-3	-1	9	1	3
67	68	-1	0	1	0	0
66	68	-2	0	4	0	0
71	70	3	2	9	4	6
67	64	-1	-4	1	16	4
70	67	2	-1	4	1	-2
68	72	0	4	0	16	0
69	70	1	2	1	4	2
	Total	-1	2	29	42	13

$$\begin{aligned} r_{XY} = r_{UV} &= \frac{n \sum uv - \sum u \cdot \sum v}{\sqrt{\{n \sum u^2 - (\sum u)^2\} \{n \sum v^2 - (\sum v)^2\}}} \\ &= \frac{8 \times 13 - (-1) \times 2}{\sqrt{(8 \times 29 - 1)(8 \times 42 - 4)}} = \frac{106}{\sqrt{231 \times 308}} \approx 0.3974 \end{aligned}$$

Exercise:

Find the coefficient of correlation between X and Y using the following data:

X :	5	10	15	20	25
Y :	16	19	23	26	30

Rank Correlation Coefficient

Sometimes the actual numerical values of X and Y may not be available, but the positions of the actual values arranged in order of merit (ranks) only may be available. The ranks of X and Y will in general, be different and hence may be considered as random variables. Let them be denoted by U and V . The correlation coefficient between U and V is called *the rank correlation coefficient* between (the ranks of) X , Y and denoted by ρ_{XY} .

Let us now derive a formula for ρ_{XY} or r_{UV} . Since U represents ranks of n values of X , U takes the values $1, 2, 3, \dots, n$.

Similarly V takes the same values $1, 2, 3, \dots, n$ in a different order.

$$D = U - V$$

$$\rho_{XY} = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

[Note: The formula for the rank correlation coefficient is known as *spearman's formula*. The values of r_{XY} and ρ_{XY} (or r_{UV}) will be, in general, different.

Example:

Ten students got the following percentage of marks in Mathematics and Physical sciences:

Students:	1	2	3	4	5	6	7	8	9	10
Marks in Mathematics:	78	36	98	25	75	82	90	62	65	39
Marks in Phy. Sciences:	84	51	91	60	68	62	86	58	63	47

Calculate the rank correlation coefficient.

Solution:

Denoting the ranks in Mathematics and in Phy. Sciences by U and V , we have the following values of U and V :

U :	4	9	1	10	5	3	2	7	6	8
V :	3	9	1	7	4	6	2	8	5	10
D :	1	0	0	3	1	-3	0	-1	1	-2
D^2 :	1	0	0	9	1	9	0	1	1	4

$\therefore \Sigma d^2 = 26$

$$\begin{aligned}\rho_{XY} = r_{UV} &= 1 - \frac{6 \Sigma d^2}{n(n^2 - 1)} \\ &= 1 - \frac{6 \times 26}{10 \times 99} = 0.8424\end{aligned}$$

Exercise:

Ten competitors in a beauty contest were ranked by three judges as follows:

	Competitors									
Judges	1	2	3	4	5	6	7	8	9	10
A:	6	5	3	10	2	4	9	7	8	1
B:	5	8	4	7	10	2	1	6	9	3
C:	4	9	8	1	2	3	10	5	7	6

Discuss which pair of judges have the nearest approach to common taste of beauty.

Regression

When the random variables X and Y are linearly correlated, the points plotted on the scatter diagram, corresponding to n pairs of observed values of X and Y , will have a tendency to cluster round a straight line. This straight is called *the regression line*. The regression line can be taken as the best fitting straight line for the observed pairs of values of X and Y in the least square sense, with which the students are familiar.

When two R.V.'s X and Y are linearly correlated, we may not know which variable takes independent values. If we treat X as the independent variable and hence assume that the values of Y depend on those of X , the regression line is called the *regression line of Y on X* . If we assume that the values of X depend on those of the independent variable Y , the *regression line of X on Y* is obtained. Thus in situations where the distinction cannot be made between the R.V.'s X and Y as to which is the independent variable and which is the dependent variable, there will be two regression lines. However, when the value of $Y(X)$ is to be predicted corresponding to a specified value of $X(Y)$, we should make use of the regression line of $Y(X)$ on $X(Y)$.

Equation of the Regression Line of Y on X:

By the principle of least squares, the normal equations which give the values of a and b .

are
$$\sum y_i = a \sum x_i + nb \quad (2)$$

and
$$\sum x_i y_i = a \sum x_i^2 + b \sum x_i \quad (3)$$

Dividing equation (2) by n , we get

$$\bar{y} = a \bar{x} + b \quad (4)$$

the equation of the regression line of Y on X as

$$y - \bar{y} = \frac{p_{XY}}{\sigma_X^2} (x - \bar{x})$$

$$y - \bar{y} = \frac{r_{XY} \sigma_Y}{\sigma_X} (x - \bar{x})$$

$$\left[\because r_{XY} = \frac{p_{XY}}{\sigma_X \sigma_Y} \right]$$

Equation of the Regression Line of X on Y:

In a similar manner, assuming the equation of the regression line of X and Y as $x = ay + b$ and using the equations

we can get the equation of the regression line of X on Y as

$$x - \bar{x} = \frac{P_{XY}}{\sigma_Y^2} (y - \bar{y})$$

or

$$x - \bar{x} = \frac{r_{XY} \sigma_X}{\sigma_Y} (y - \bar{y})$$

$$X_{on Y}; (y - \bar{y}) = \frac{r_{XY} \sigma_Y}{\sigma_X} (x - \bar{x})$$

$$X_{on Y}; (x - \bar{x}) = \frac{r_{XY} \sigma_X}{\sigma_Y} (y - \bar{y})$$

Example:

Obtain the equations of the lines of regression from the following data:

X:	1	2	3	4	5	6	7
Y:	9	8	10	12	11	13	14

Example:

Obtain the equations of the lines of regression from the following data:

X:	1	2	3	4	5	6	7
Y:	9	8	10	12	11	13	14

Find $\bar{y}$, when $\bar{x} = 3.4$
 $y = 10.9$
 $\bar{y} = 3.4$

x_i y_i x_i^2 y_i^2 $x_i y_i$

1
2
3
4
5
6
7

Total $\sum x_i$ $\sum y_i$ $\sum x_i^2$ $\sum y_i^2$ $\sum x_i y_i$

$$\bar{x} = \frac{\sum x_i}{n} \quad \sigma_x = \sqrt{\frac{1}{n} \sum x_i^2 - \left(\frac{\sum x_i}{n}\right)^2}$$

$$\bar{y} = \frac{\sum y_i}{n} \quad \sigma_y = \sqrt{\frac{1}{n} \sum y_i^2 - \left(\frac{\sum y_i}{n}\right)^2}$$

$$\text{Cov}(X, Y) = \frac{1}{n} \sum x_i y_i - \frac{1}{n} \sum x_i \cdot \frac{1}{n} \sum y_i$$

$$r_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y}$$

Regression line of Y on X

$$(y - \bar{y}) = \frac{r_{xy} \sigma_y}{\sigma_x} (x - \bar{x})$$

$y = a + bx$ —→ ①
 Regression of X on Y

$$(x - \bar{x}) = \frac{r_{xy} \sigma_x}{\sigma_y} (y - \bar{y})$$

$x = a + by$ —→ ②
 10.9

Solution:

X	Y	U = X - 4	V = Y - 11	U ²	V ²	UV
1	9	-3	-2	9	4	6
2	8	-2	-3	4	9	6
3	10	-1	-1	1	1	1
4	12	0	1	0	1	0
5	11	1	0	1	0	0
6	13	2	2	4	4	4
7	14	3	3	9	9	9
Total		0	0	28	28	26

$$\bar{x} = E(X) = 4 + \frac{1}{n} \sum u = 4$$

$$\bar{y} = E(Y) = 11 + \frac{1}{n} \sum v = 11$$

$$\sigma_X^2 = \frac{1}{n} \sum u^2 - \left(\frac{1}{n} \sum u \right)^2 = \frac{1}{7} \times 28 = 4$$

$$\sigma_Y^2 = \frac{1}{n} \sum v^2 - \left(\frac{1}{n} \sum v \right)^2 = \frac{1}{7} \times 28 = 4$$

$$C_{XY} = \frac{1}{n} \sum uv - \left(\frac{1}{n} \sum u \right) \left(\frac{1}{n} \sum v \right) = \frac{1}{7} \times 26 = 3.7$$

The regression line of Y on X is

$$y - \bar{y} = \frac{P_{XY}}{\sigma_X^2} (x - \bar{x})$$

i.e., $y - 11 = \frac{3.7}{4} (x - 4)$

i.e., $3.7x - 4y + 29.2 = 0$

The regression line of X on Y is

$$x - \bar{x} = \frac{P_{XY}}{\sigma_Y^2} (y - \bar{y})$$

i.e., $x - 4 = \frac{3.7}{4} (y - 11)$

i.e., $4x - 3.7y + 24.7 = 0$

Exercise:

Find the equations of the regression lines from the following data. Also estimate the value of Y when $X = 71$ and the value of X when $Y = 70$.

$X:$	65	66	67	67	68	69	70	72
$Y:$	67	68	65	68	72	72	69	71

Exercise:

Obtain the equations of the regression lines from the following data, using the method of least squares.

Also estimate the value of (i) Y , when $X = 38$ and (ii) X , when $Y = 18$.

$X:$	22	26	29	30	31	31	34	35
$Y:$	20	20	21	29	27	24	27	31

$(X_1, X_2, X_3) \rightarrow$ Trivariate Distribution

$$r(X_1, X_2) = r_{X_1 X_2} = r_{12} = r_{21}$$

$$r(X_1, X_3) = r_{X_1 X_3} = r_{13} = r_{31}$$

$$r(X_2, X_3) = r_{X_2 X_3} = r_{23} = r_{32}$$

$$r_{x_1} = \frac{\text{Cov}(x_1, x)}{\sigma_x \cdot \sigma_x} = r_{yx}$$

Multiple and Partial Correlation

Multiple Correlation

Suppose one variable may be influenced by many other variables. Such a correlation is called multiple correlation.

Multiple Correlation Co-efficient (R)

In a trivariate distribution (X_1, X_2, X_3) , the multiple correlation co-efficient of X_1 on X_2 & X_3 is denoted and defined as

$$R_{1.23}^2 = \frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}$$

||| $R_{2.13}$ & $R_{3.12}$

Note:

$$* R_{1.23}^2 \leq 1$$

$$|r_{xy}| \leq 1$$

$$* 0 \leq R_{1.23} \leq 1$$

Partial Correlation

The correlation between 2 Variables X_1 and X_2 may be partly due to the correlation of a third Variable X_3 with both X_1 and X_2 . In such a situation, the effect of X_3 on each of X_1 and X_2 were eliminated. Such a correlation is called Partial Correlation.

Partial Correlation Co-efficient:

The partial correlation co-efficient between X_1 & X_2 after eliminating the linear effect of X_3 , is denoted & defined as

$$r_{12.3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}}$$

$$r_{13.2} = \frac{r_{13} - r_{12} r_{32}}{\sqrt{(1 - r_{12}^2)(1 - r_{32}^2)}} \quad \text{and} \quad r_{23.1} = \frac{r_{23} - r_{21} r_{31}}{\sqrt{(1 - r_{21}^2)(1 - r_{31}^2)}}$$

Example:

In a trivariate distribution : $r_{12} = 0.77$, $r_{13} = 0.72$ and $r_{23} = 0.52$

Find the the partial correlation coefficient $r_{12.3}$ and multiple correlation coefficient $R_{1.23}$.

Solution:

$$r_{12.3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}} = \frac{0.77 - 0.72 \times 0.52}{\sqrt{[1 - (0.72)^2][1 - (0.52)^2]}} = 0.62$$

$$\begin{aligned} R_{1.23}^2 &= \frac{r_{12}^2 + r_{13}^2 - 2r_{12} r_{13} r_{23}}{1 - r_{23}^2} \\ &= \frac{(0.77)^2 + (0.72)^2 - 2 \times 0.77 \times 0.72 \times 0.52}{1 - (0.52)^2} = 0.7334 \end{aligned}$$

$$\therefore R_{1.23} = \pm 0.8564$$

Exercise:

In a trivariate distribution : $r_{12} = 0.7, r_{23} = r_{31} = 0.5$.

Find (i) $r_{23.1}$, (ii) $R_{1.23}$,

Multiple Regression

Method of Least Squares

$$y, x_1, x_2$$

$$\begin{aligned} \sum_{i=1}^n y_i &= a \cdot \sum x_{1i} + b \cdot \sum x_{2i} + n \cdot c \\ \sum x_{1i} y_i &= a \cdot \sum x_{1i}^2 + b \cdot \sum x_{1i} x_{2i} + c \cdot \sum x_{1i} \\ \sum x_{2i} y_i &= a \cdot \sum x_{1i} x_{2i} + b \cdot \sum x_{2i}^2 + c \cdot \sum x_{2i} \end{aligned}$$

Multiple Regression

Normal Estimation
Equations for
Multiple Linear
Regression

$$\begin{array}{ccccccc}
 nb_0 + b_1 \sum_{i=1}^n x_{1i} & + & b_2 \sum_{i=1}^n x_{2i} & + & \cdots & + & b_k \sum_{i=1}^n x_{ki} & = & \sum_{i=1}^n y_i \\
 b_0 \sum_{i=1}^n x_{1i} + b_1 \sum_{i=1}^n x_{1i}^2 & + & b_2 \sum_{i=1}^n x_{1i}x_{2i} & + & \cdots & + & b_k \sum_{i=1}^n x_{1i}x_{ki} & = & \sum_{i=1}^n x_{1i}y_i \\
 \vdots & & \vdots & & \vdots & & \vdots & & \vdots \\
 b_0 \sum_{i=1}^n x_{ki} + b_1 \sum_{i=1}^n x_{ki}x_{1i} & + & b_2 \sum_{i=1}^n x_{ki}x_{2i} & + & \cdots & + & b_k \sum_{i=1}^n x_{ki}^2 & = & \sum_{i=1}^n x_{ki}y_i
 \end{array}$$

Example:

The following data represent the chemistry grades for a random sample of 12 freshmen at a certain college along with their scores on an intelligence test administered while they were still seniors in high school. The number of class periods missed is also given.

Student	Chemistry Grade, y	Test Score, x_1	Classes Missed, x_2
1	85	65	1
2	74	50	7
3	76	55	5
4	90	65	2
5	85	55	6
6	87	70	3
7	94	65	2
8	98	70	5
9	81	55	4
10	91	70	3
11	76	50	1
12	74	55	4

- (a) Fit a multiple linear regression equation of the form $\hat{y} = b_0 + b_1x_1 + b_2x_2$.
- (b) Estimate the chemistry grade for a student who has an intelligence test score of 60 and missed 4 classes.

Multiple Regression

Regression equation of X_1 on X_2 and X_3 is given by

$$(X_1 - \bar{X}_1) \frac{\omega_{11}}{\sigma_1} + (X_2 - \bar{X}_2) \frac{\omega_{12}}{\sigma_2} + (X_3 - \bar{X}_3) \frac{\omega_{13}}{\sigma_3} = 0$$

where $\omega = \begin{vmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{vmatrix}$

$$\omega_{11} = \begin{vmatrix} 1 & r_{23} \\ r_{32} & 1 \end{vmatrix} = 1 - r_{23}^2$$

$$\omega_{12} = - \begin{vmatrix} r_{21} & r_{23} \\ r_{31} & 1 \end{vmatrix} = r_{13} r_{23} - r_{21}$$

$$\omega_{13} = r_{23} r_{12} - r_{13}$$

Example:

Find the regression equation of X_1 on X_2 and X_3 given the following results :—

Trait	Mean	Standard deviation	r_{12}	r_{23}	r_{31}
X_1	28.02	4.42	+ 0.80	—	—
X_2	4.91	1.10	—	-0.56	—
X_3	594	85	—	—	- 0.40

**where X_1 = Seed per acre; X_2 = Rainfall in inches
 X_3 = Accumulated temperature above 42°F.**

Solution. Regression equation of X_1 on X_2 and X_3 is given by

$$(X_1 - \bar{X}_1) \frac{\omega_{11}}{\sigma_1} + (X_2 - \bar{X}_2) \frac{\omega_{12}}{\sigma_2} + (X_3 - \bar{X}_3) \frac{\omega_{13}}{\sigma_3} = 0$$

where $\omega = \begin{vmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{vmatrix}$

$$\omega_{11} = \begin{vmatrix} 1 & r_{23} \\ r_{32} & 1 \end{vmatrix} = 1 - r_{23}^2 = 1 - (-0.56)^2 = 0.686$$

$$\omega_{12} = - \begin{vmatrix} r_{21} & r_{23} \\ r_{31} & 1 \end{vmatrix} = r_{13} r_{23} - r_{21} = -0.576$$

$$\omega_{13} = r_{23} r_{12} - r_{13} = (-0.56)(0.80) - (-0.40) = -0.048$$

$\therefore$ Required equation of plane of regression of X_1 on X_2 and X_3 is given by

$$\frac{0.686}{4.42} (X_1 - 28.02) + \frac{(-0.576)}{1.10} (X_2 - 4.91) + \frac{(-0.048)}{85.00} (X_3 - 594) = 0$$

